1. Smart Hostel Management System

✅ Final Essential + Important Features (Optimized)

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1. 🔐 Authentication & User Management

- Role-based login (Student, Staff, Admin)
- Student ID / Username & Password login
- Google / Microsoft OAuth login *(optional)*
- Forgot password & reset flow
- Login history & device tracking

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2. 👨‍🎓 Student Module

- Profile management
- Hostel application & approval status
- Room & bed details
- Fee details & **receipt download**
- Complaint registration with **image upload**
- Complaint status tracking
- Gate pass / outing requests
- Attendance view
- **Notifications (important updates, approvals, dues)**
- **Emergency SOS feature**

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3. 🛠️ Admin / Warden Module

- Hostel structure (Block, Room, Bed management)
- Room allocation (manual)
- Student approval, transfer & checkout
- Fee structure management
- Staff management
- Announcements / notices
- **Task dashboard (complaints, maintenance tracking)**
- **Analytics dashboard (occupancy, complaints, revenue)**

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4. 👷 Staff Module

- Staff login
- **Task dashboard (assigned complaints)**
- Complaint handling & status updates
- Maintenance updates
- Attendance marking

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5. 🍽️ Mess Management System

- Daily / weekly menu management
- Meal booking / cancellation
- Mess fee calculation

- **Food feedback & ratings**
- **Monthly mess consumption report**

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6. Complaint & Maintenance System

- Category-based complaints
- Priority-based handling
- **Image & note attachments**
- Complaint status tracking
- Maintenance records

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7. Visitor & Gate Pass Management

- Visitor entry records
- Student gate pass / outing requests
- Visitor logs

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8. Fee & Payment Management

- Fee payment tracking
- Due / pending fee tracking
- **Receipt generation & download**
- Basic fee reports

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9. Reports & Export

- Occupancy reports
- Fee/payment reports
- Complaint reports
- **Export to PDF / Excel**

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10. Smart Features

- **QR / RFID-based attendance integration**